



## Modelling Financial Markets

### Seminar 8: Backtesting and Evaluating Risk Forecasts — Hands-on



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## Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CLT: Central Limit Theorem
- ▣ DGP: Data-Generating Process
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ EGARCH: Exponential GARCH
- ▣ ES: Expected Shortfall
- ▣ ESR: Expected Shortfall Regression (backtest)
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ LR: Likelihood Ratio (test)
- ▣ LRV: Long-Run Variance
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ POF: Proportion Of Failures (test)

## Acronyms used in this material (2/2)

- **US: United States**
- **UTC: Coordinated Universal Time (Temps Universel Coordonné)**

## Today's Question and Required Outputs

- Question of the seminar: **which forecasting procedure would you provisionally choose for VaR 1% and ES 2.5%, and how strong is the evidence?**
- Route
  - ▶ Step 1: A1, A3 (counting and timing breaches on paper), A5, A7 (traffic lights for VaR and ES), B1 (six VaR models on the S&P 500)
  - ▶ Step 2: B3 (ES by bootstrap and simulation), B5 (comparing forecasts with HAC standard errors), B7 (conformal VaR in crises), C2 (critique of an AI answer); B2 at home
- Required outputs (checklist on the last core slide): the A answers, one table and one figure per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
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## Exercise Map

| Exercise   | Question                                                                                 | Type     | Model              |
|------------|------------------------------------------------------------------------------------------|----------|--------------------|
| A1         | is 7 breaches in 250 days evidence against a VaR 1%?                                     | Solved   | –                  |
| A3         | do the BET breaches of 2007–2010 cluster?                                                | Solved   | –                  |
| A5, A7     | traffic lights for VaR counts and for ES ( $Z_2$ )                                       | Solved   | –                  |
| B1         | which of six VaR models passes on the S&P 500?                                           | Solved   | –                  |
| B3, B5     | is ES underestimated? is FHS significantly better?                                       | Solved   | –                  |
| B7         | does conformal VaR keep its coverage in crises?                                          | Solved   | –                  |
| C2         | what is wrong in an AI answer?                                                           | Proposed | A3, A5             |
| B2         | would a supervisor have penalised GARCH-t on the BET?                                    | Proposed | A5, B1             |
| Extensions | A1 Extended, A2, A4, A6, A8–A10, B4, B6, B8, B9, C1, C3<br>(optional, after the seminar) | both     | named on each task |

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: verification and inference on real forecasts, each ending with an interpretation question; Part C: open questions and AI critique

## Data Manifest and the Forecast Panel

| Market  | Source, field      | Calendar              | Evaluation days              |
|---------|--------------------|-----------------------|------------------------------|
| S&P 500 | GSPC.INDX, close   | US trading days       | 03.01.2007–18.09.2026, 4 959 |
| BET     | BET, close         | BVB trading days      | 03.01.2007–18.09.2026, 4 937 |
| Bitcoin | BTC-USD.CC, close  | every day, 24:00 UTC  | 14.06.2017–18.09.2026, 3 384 |
| EUR/RON | BNR reference rate | Romanian working days | 11.06.2009–18.09.2026, 4 352 |

- Market files: `data/market/<SYMBOL>.csv`; losses  $L_t = -\ln(P_t/P_{t-1})$  in **decimals** ( $0.01 = 1\%$ ), each series on its own calendar, no filling of missing days
- Forecast panel `fc(market)[0][model]`, one row per evaluation day: L, VaR1, VaR2.5, VaR5, ES2.5, scale ( $\sigma_t$ ), mu, pit; the six models of Lecture 8, 1000-day rolling window, refit every 20 days, data up to  $t - 1$  only
- The panel is supplied: in class you test forecasts, you do not re-estimate models (optional: the estimation code in the notebook)

## Setup: Run and Verify

```
# section "Common helpers" defines fc,  
# hits, kupiec, christoffersen,  
# duration_test, dq_test, fz0, ...  
F, tails = fc('sp500')  
g = F['GARCH-t']  
print(len(g), g.index[0], g.index[-1])  
# loss, VaR 1%, VaR 2.5%, ES 2.5%, scale  
print(g.iloc[:3, [-1, 0, 1, 3, 6]])
```

- Expected output: 4959 rows, 03.01.2007 to 18.09.2026
- A negative L is a gain; VaR and ES are positive loss thresholds
- If your count or first row differs, stop and check the symbol, the start date and the model name

| date       | L         | VaR1     | VaR2.5   | ES2.5    | scale    |
|------------|-----------|----------|----------|----------|----------|
| 2007-01-03 | 0.001199  | 0.012031 | 0.009944 | 0.012132 | 0.005298 |
| 2007-01-04 | -0.001228 | 0.011894 | 0.009830 | 0.011995 | 0.005241 |
| 2007-01-05 | 0.006103  | 0.011745 | 0.009706 | 0.011844 | 0.005178 |

## Notation, and What Is Known When a Forecast Is Issued

- ▣ Loss  $L_t = -r_t$ ; forecasts  $\text{VaR}_t$ ,  $\text{ES}_t$ ,  $\sigma_t$  are issued at the close of day  $t - 1$  and use only  $\mathcal{F}_{t-1}$  (returns up to  $r_{t-1}$ )
- ▣ Level = tail probability  $\alpha$ : VaR 1% ( $\alpha = 0.01$ ), VaR 2.5% and ES 2.5% ( $\alpha = 0.025$ ); breach  $I_t = 1\{L_t > \text{VaR}_t\}$ , target rate  $\alpha$
- ▣  $T$ : evaluation days;  $x = \sum_t I_t$ : breaches;  $\hat{\pi} = x/T$ ;  $n_{ij}$ : days in state  $j$  after a day in state  $i$
- ▣  $H_0$  of every backtest:  $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$ , i.e. the  $I_t$  are i.i.d. Bernoulli( $\alpha$ ); a test checks one implication (count, memory, spacing)
- ▣ Scores for comparing forecasts: pinball for VaR, FZ0 for (VaR, ES); lower is better
- ▣ One worked line:  $T = 250$ ,  $x = 7$ :  $\hat{\pi} = 2.8\%$  against  $\alpha = 1\%$ , 2.5 breaches expected

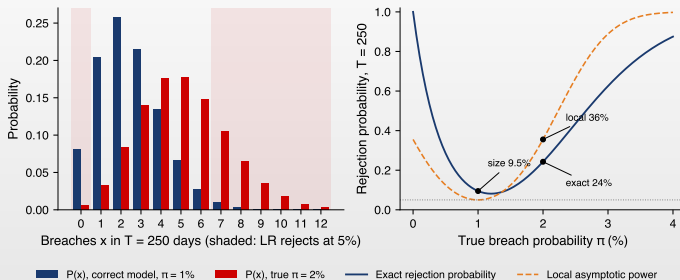
## A1 [Solved]: Kupiec from a Breach Count — Task

- ▣ **Question:** is  $x = 7$  breaches in  $T = 250$  days evidence against a VaR 1% model?
- ▣ **Inputs:**  $T = 250$ ,  $x = 7$ ,  $\alpha = 0.01$ ; the hits are i.i.d. Bernoulli( $\pi$ ); no data file
- ▣ **Tasks**
  1. Compute  $\hat{\pi} = x/T$  and the log-likelihoods  $\ell(\alpha) = (T - x) \ln(1 - \alpha) + x \ln \alpha$  (restricted) and  $\ell(\hat{\pi})$  (unrestricted).
  2. Compute  $\text{LR}_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)]$ , its  $\chi^2_1$   $p$ -value, and the exact one-sided probability  $P(X \geq 7)$  under Binomial(250, 0.01).
  3. List the integers  $x$  with  $\text{LR}_{uc}(x) \leq 3.84$ : the acceptance region of the rule “reject if  $\text{LR}_{uc} > 3.84$ ”.
  4. Sum Binomial probabilities outside that region to get the actual size ( $\pi = 1\%$ ) and the power against  $\pi = 2\%$ ; compare with the local power  $P(\chi^2_1(\lambda) > 3.84)$ ,  $\lambda = T(\pi - \alpha)^2/(\alpha(1 - \alpha))$ .
- ▣ **Report:**  $\hat{\pi}$ , the two log-likelihoods,  $\text{LR}_{uc}$  with both  $p$ -values, the acceptance region, nominal size, actual size and the two powers
- ▣ **Interpretation:** can one year of VaR 1% data detect a model that doubles the breach rate?
- ▣ **Notebook:** section A1

## A1 [Solved]: Solution (1/2) — The Test Step by Step

1.  $\hat{\pi} = 7/250 = 0.028$ ;  $\ell(\alpha) = 243 \ln 0.99 + 7 \ln 0.01 = -34.678$ ;  $\ell(\hat{\pi}) = 243 \ln 0.972 + 7 \ln 0.028 = -31.930$ .
  2.  $LR_{uc} = 2(-31.930 - (-34.678)) = 5.50$ ;  $\chi_1^2$   $p = 0.019$ ; exact  $P(X \geq 7) = 0.014$ : both reject at 5% (too many breaches).
  3. Evaluating  $LR_{uc}(x)$  for  $x = 0, 1, 2, \dots$ :  $LR_{uc}(0) = -500 \ln 0.99 = 5.03 > 3.84$ ,  $LR_{uc}(6) = 3.56$ ,  $LR_{uc}(7) = 5.50$ : the acceptance region is  $\{1, \dots, 6\}$ .
  4. Three different quantities: **nominal size** 5% (the  $\chi_1^2$  critical value), **actual size** of this rule (next slide), **power** against  $\pi = 2\%$  (next slide).
- ☐ Interpretation: 7 breaches are 2.8 times the 2.5 expected; a correct VaR 1% produces 7 or more breaches in only 0.014 of years

## A1 [Solved]: Solution (2/2) — Size and Power



- Actual size:  $P(X = 0) + P(X \geq 7) = 8.11\% + 1.37\% = 9.5\%$  under  $\pi = 1\%$ , not 5%
- Exact power at  $\pi = 2\%$ :  $1 - P(1 \leq X \leq 6) = 1 - 75.7\% = 24\%$
- Local:  $\lambda = 250 \cdot 0.01^2 / 0.0099 = 2.53$ ,  $P(\chi_1^2(\lambda) > 3.84) = 36\%$ : the asymptotics overstate the power
- Interpretation: with 2.5 expected breaches the lattice of  $x$  dominates; a model that doubles the breach rate passes one year of testing about three times in four

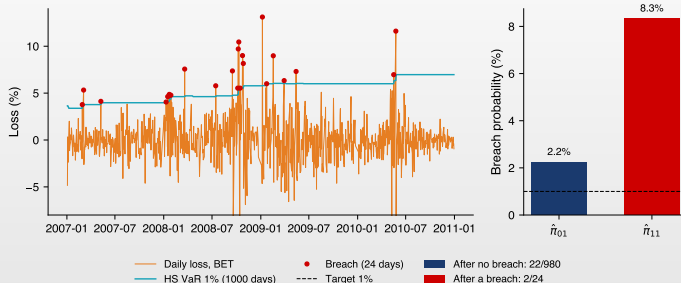
## A3 [Solved]: From a Hit Sequence to Transitions — Task

- **Question:** do the breaches of historical simulation on the BET in 2007–2010 cluster, or are there simply too many?
- **Inputs**
  - ▶ Teaching sequence of 12 days:  $I_t = 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0$
  - ▶ BET (data/market/BET.csv), HS VaR 1% (1000-day window), 2007–2010:  $T = 1005$  days, so  $n = T - 1 = 1004$  transitions (the first day is conditioned on)
  - ▶ Transition counts:  $n_{00} = 958$ ,  $n_{01} = 22$ ,  $n_{10} = 22$ ,  $n_{11} = 2$
- **Tasks**
  1. Count the 11 transitions of the 12-day sequence into a  $2 \times 2$  table  $(n_{00}, n_{01}, n_{10}, n_{11})$ .
  2. For the BET, compute  $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$ ,  $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$  and  $\hat{\pi} = (n_{01} + n_{11})/n$ .
  3. Write the restricted log-likelihood  $\ln L(\hat{\pi}, \hat{\pi})$  and the unrestricted  $\ln L(\hat{\pi}_{01}, \hat{\pi}_{11})$ ; compute  $\text{LR}_{ind}$  and its  $\chi^2_1$   $p$ -value.
  4. Compute  $\text{LR}_{uc}$  on the same  $n$  transitions ( $x = 24$ ), then  $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind}$  and its  $\chi^2_2$   $p$ -value.
- **Report:** the toy table, three probabilities,  $\text{LR}_{ind}$ ,  $\text{LR}_{uc}$ ,  $\text{LR}_{cc}$  with  $p$ -values
- **Interpretation:** does HS fail because of too many breaches or because breaches cluster?
- Notebook: section A3

## A3 [Solved]: Solution (1/2) — Counts and Likelihoods

1. Toy sequence: pairs  $(l_{t-1}, l_t)$  give  $n_{00} = 6$ ,  $n_{01} = 2$ ,  $n_{10} = 2$ ,  $n_{11} = 1$  (total 11).
  2. BET:  $\hat{\pi}_{01} = 22/(958 + 22) = 0.0224$ ;  $\hat{\pi}_{11} = 2/(22 + 2) = 0.0833$ ;  $\hat{\pi} = (22 + 2)/1004 = 0.0239$ .
  3.  $\ln L(\pi_{01}, \pi_{11}) = n_{00} \ln(1 - \pi_{01}) + n_{01} \ln \pi_{01} + n_{10} \ln(1 - \pi_{11}) + n_{11} \ln \pi_{11}$ ; each part is a Bernoulli likelihood, so the MLEs are the ratios of step 2;  $\text{LR}_{ind} = 2.32$ ,  $p = 0.128$ : independence not rejected.
  4.  $\text{LR}_{uc} = 14.11$  on the 1004 transitions;  $\text{LR}_{cc} = 14.11 + 2.32 = 16.43$ ,  $p = 0.0003$  ( $\chi^2_2$ : both  $\pi_{01}$  and  $\pi_{11}$  fixed at  $\alpha$ ): rejected.
- On the same transitions the decomposition  $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind}$  is exact; with  $\text{LR}_{uc}$  on all  $T$  days it holds only asymptotically

## A3 [Solved]: Solution (2/2) — What the Chart Shows



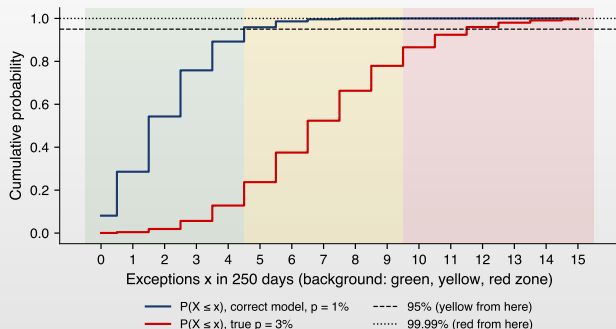
- The HS VaR moves in steps and stays flat while losses cluster in late 2008: 24 breaches where 10 are expected
- After a breach the breach probability is almost four times higher (0.0833 versus 0.0224), yet it rests on only 2 consecutive pairs
- Interpretation: HS fails mainly through too many breaches ( $\hat{\pi} \approx 2.4\%$ ); the clustering is visible but four years of data cannot confirm it with a first-order test

## A5 [Solved]: The Traffic Light from the Binomial — Task

- **Question:** a bank reports 6 exceptions of its VaR 1% in 250 days: which zone, what capital consequence, and how unusual is it?
- **Inputs**
  - ▶  $X \sim \text{Binomial}(250, 0.01)$  for a correct model; green while  $P(X \leq x) < 95\%$ , red from  $P(X \leq x) \geq 99.99\%$  (BCBS (1996))
  - ▶ Lookup table (right): 1996 plus factor on the multiplier 3; FRTB bank-wide multiplier (BCBS (2019))
- **Tasks**
  1. Compute  $P(X \leq x)$  for  $x = 4, 5, 9, 10$  and find where the zones start.
  2. For  $x = 6$ : give the zone, the 1996 multiplier, the FRTB multiplier and  $P(X \geq 6) = 1 - P(X \leq 5)$ .
  3. Compute  $P(X \leq 4)$  when the true breach probability is 3%.
  4. Compute the probability that a correct model leaves green,  $P(X \geq 5)$ , and compare with reaching red.
- **Report:** four probabilities, the zones, the answers to 2–4
- **Interpretation:** which error does the traffic light control, and which does it leave large?
- **Notebook:** section A5

| Exceptions | 1996: plus factor | FRTB multiplier |
|------------|-------------------|-----------------|
| 0–4        | 0                 | 1.50            |
| 5          | 0.40              | 1.70            |
| 6          | 0.50              | 1.76            |
| 7          | 0.65              | 1.83            |
| 8          | 0.75              | 1.88            |
| 9          | 0.85              | 1.92            |
| 10 or more | 1.00              | 2.00            |

## A5 [Solved]: Solution



- $P(X \leq 4) = 89.22\%$ ,  $P(X \leq 5) = 95.88\%$ ,  
 $P(X \leq 9) = 99.97\%$ ,  $P(X \leq 10) = 99.99\%$ :  
 green 0–4, yellow 5–9, red 10 or more
- $x = 6$ : yellow; 1996 multiplier  $3 + 0.50 = 3.50$ ;  
 FRTB 1.76;  
 $P(X \geq 6) = 100\% - 95.88\% = 4.12\%$
- True 3%:  $P(X \leq 4) = 12.8\%$ : a model that  
 triples the breach rate stays green about one  
 year in eight
- A correct model leaves green with  
 $P(X \geq 5) = 10.78\%$  but almost never reaches  
 red: the zones control false red alarms; the  
 zone is a lookup rule, not a 5% significance test

A7 [Solved]:  $Z_1$  and  $Z_2$  on One Year — Task

□ **Question:** do eight breach days in one year show that ES 2.5% is underestimated?

□ **Inputs**

- ▶  $T = 250$  days,  $\alpha = 0.025$ , so  $T\alpha = 6.25$ ; the table lists the eight days that are **already identified** as breaches of VaR 2.5%
- ▶  $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{ES_t}$ ,  $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t I_t}{ES_t}$  (Acerbi & Szekely (2014))

□ **Tasks**

1. Complete the column  $L_t/ES_t$  and compute each day's contribution  $L_t/(ES_t T\alpha)$  to the sum in  $Z_2$ .
2. Compute  $Z_1$  and  $Z_2$ ; classify  $Z_2$  with the reference thresholds  $-0.70$  and  $-1.8$  (illustrative; B3 calibrates them for a model).
3. Replace one non-breach day of the same window by a breach with  $L = 7\%$ ,  $ES = 3.5\%$  ( $T$  stays 250) and recompute  $Z_2$ .
4. Show  $E[L_t I_t | \mathcal{F}_{t-1}] = \alpha ES_t$  for a correct continuous forecast and deduce  $E[Z_2] = 0$ .

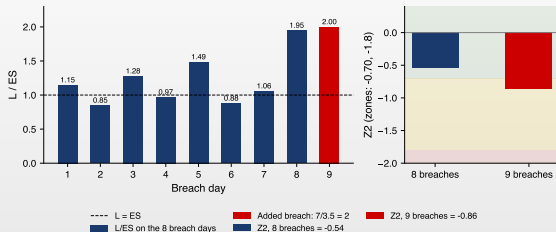
□ **Report:** the ratios,  $Z_1$ ,  $Z_2$  before and after, the zone, the proof

□ **Interpretation:** which days drive  $Z_2$ ?

□ **Notebook:** section A7

| Breach day | $L_t$ (%) | $ES_t$ (%) | $L_t/ES_t$ |
|------------|-----------|------------|------------|
| 1          | 3.9       | 3.4        | ...        |
| 2          | 2.8       | 3.3        | ...        |
| 3          | 4.6       | 3.6        | ...        |
| 4          | 3.1       | 3.2        | ...        |
| 5          | 5.2       | 3.5        | ...        |
| 6          | 2.9       | 3.3        | ...        |
| 7          | 3.6       | 3.4        | ...        |
| 8          | 7.4       | 3.8        | ...        |

## A7 [Solved]: Solution



- Ratios: 1.147, 0.848, 1.278, 0.969, 1.486, 0.879, 1.059, 1.947; sum 9.613; contributions = ratio/6.25, from 0.136 (day 2) to 0.312 (day 8)
- $Z_1 = 1 - 9.613/8 = -0.202$ : tail losses 20% above ES on average;  $Z_2 = 1 - 9.613/6.25 = -0.538 > -0.70$ : green
- Added breach, ratio 2:  $Z_2 = -0.86 < -0.70$ : yellow
- Proof:  $E[L_t | \mathcal{F}_{t-1}] = P(L_t > \text{VaR}_t | \mathcal{F}_{t-1}) E[L_t | L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = \alpha \text{ES}_t$ ;  $T\alpha$  is constant, so  $E[Z_2] = 0$  ( $Z_1$  divides by the random  $N_T$ )
- Interpretation:  $Z_2$  reacts to both the number and the size of breaches; one large loss can change the zone

## B1 [Solved]: Six VaR Models on the S&P 500 — Task

- **Question:** which of the six procedures pass the VaR 1% backtests, and which test decides?

- **Inputs**

- ▶ Notebook section B1; panel `fc('sp500')`, L, VaR1;  $T = 4\,959$ ; table: ten GARCH-t days, September 2008
- ▶ Helpers: `kupiec`, `christoffersen`, `duration_test` (end durations censored), `dq_test` (constant, 4 lagged hits,  $\text{VaR}_t$ )

- **Tasks**

1. Fill the column  $I_t$  of the table and count the breaches.
2. For GARCH-t in 2008 ( $T = 253$ ): compute  $x$ ,  $\text{LR}_{uc}$ , the  $\chi^2_1$  and the exact two-sided Binomial  $p$ -values; decide at 5%.
3. For all six models on the full sample:  $x$ , rate, Kupiec (asymptotic, exact), independence, CC, Weibull shape  $b$  with the duration  $p$ -value, DQ  $p$ -value.
4. Plot the breach days and the survival curves of the durations.
5. Use the simulated size of the tests (Lecture 8) to read a DQ  $p = 0.03$  at  $T = 250$ .

- **Report:** the template table and two figures

- **Interpretation:** which models pass all tests, and why can a model with the right count still fail?

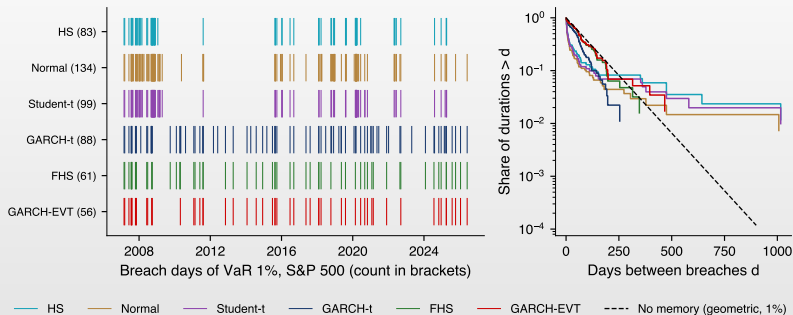
| Date       | $L_t$ (%) | $\text{VaR}_t$ (%) | $I_t$ |
|------------|-----------|--------------------|-------|
| 04.09.2008 | 3.04      | 2.98               | ?     |
| 05.09.2008 | -0.44     | 3.45               | ?     |
| 08.09.2008 | -2.03     | 3.35               | ?     |
| 09.09.2008 | 3.47      | 3.46               | ?     |
| 10.09.2008 | -0.61     | 3.99               | ?     |
| 11.09.2008 | -1.37     | 3.88               | ?     |
| 12.09.2008 | -0.21     | 3.84               | ?     |
| 15.09.2008 | 4.83      | 3.72               | ?     |
| 16.09.2008 | -1.74     | 4.68               | ?     |
| 17.09.2008 | 4.83      | 4.64               | ?     |

## B1 [Solved]: Solution (1/3) — One Model Step by Step

| Date       | $L_t$ | $\text{VaR}_t$ | $I_t$ |
|------------|-------|----------------|-------|
| 04.09.2008 | 3.04  | 2.98           | 1     |
| 05.09.2008 | -0.44 | 3.45           | 0     |
| 08.09.2008 | -2.03 | 3.35           | 0     |
| 09.09.2008 | 3.47  | 3.46           | 1     |
| 10.09.2008 | -0.61 | 3.99           | 0     |
| 11.09.2008 | -1.37 | 3.88           | 0     |
| 12.09.2008 | -0.21 | 3.84           | 0     |
| 15.09.2008 | 4.83  | 3.72           | 1     |
| 16.09.2008 | -1.74 | 4.68           | 0     |
| 17.09.2008 | 4.83  | 4.64           | 1     |

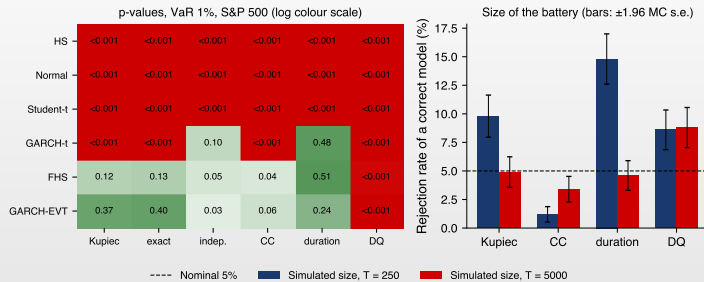
1.  $I_t = 1$  when  $L_t > \text{VaR}_t$ : 4 breaches in 10 days, against 0.1 expected: a burst, while the VaR rises only after each loss.
2. 2008:  $x = 8$  in  $T = 253$ ,  $\hat{\pi} = 3.16\%$ ;  $\text{LR}_{uc} = 7.60$ ,  $p = 0.0058$ ; exact  $p = 0.0086$ : rejected at 5%.
3. Full sample: the same hit series goes into all five tests; the counts give the rates, the transitions give independence, the gaps between breaches give the durations.
4. With 4 959 days the asymptotic and exact Kupiec  $p$ -values agree; in one year they differ (0.0058 versus 0.0086).

## B1 [Solved]: Solution (2/3) — When the Breaches Happen



- Unconditional models (HS, Normal, Student-t): bursts in 2008–2009, 2020 and 2022, then years of silence; Weibull  $b$ : HS 0.52, Normal 0.56, Student-t 0.53 (duration  $p < 0.001$ )
- Filtered models: survival curves close to the geometric line; GARCH-t 0.94 ( $p = 0.479$ ), GARCH-EVT 0.88 ( $p = 0.240$ )

## B1 [Solved]: Solution (3/3) — All Tests and Their Size



- Only GARCH-EVT passes Kupiec, CC and durations; FHS fails CC narrowly ( $p = 0.040$ ); DQ rejects all six (GARCH-EVT  $p < 0.001$ ): hits are predictable from lagged hits and the VaR level
- Size at  $T = 250$ : POF 9.8%, CC 1.2%, durations 14.8%, DQ 8.6% (Monte Carlo s.e. about 0.9 points): a DQ  $p = 0.03$  in one year is weak evidence; simulate the critical value
- Interpretation: a right count (FHS: 1.23%) can still fail on timing; only filtered models with a flexible tail survive most tests

## B3 [Solved]: Is ES Underestimated? — Task (1/2)

- **Question:** do the ES 2.5% forecasts of GARCH-t, FHS and HS understate the tail of the S&P 500?
- **Inputs**
  - ▶ Panel `fc('sp500')`:  $L$ ,  $\text{VaR}_{2.5}$ ,  $\text{ES}_{2.5}$ ,  $\text{scale}$  ( $\sigma_t$ , for HS the window standard deviation), and for HS and FHS the 25 largest window losses of each day ( $\text{tails}$ );  $T = 4\,959$
  - ▶ One day step by step, GARCH-t, 12.03.2020:  $L = 9.99\%$ ,  $\text{VaR} = 9.40\%$ ,  $\text{ES} = 13.43\%$ ,  $\sigma = 4.81\%$
- **Tasks**
  1. For that day compute the exceedance residual  $e_t = (L_t - \text{ES}_t)/\sigma_t$  and the contribution  $L_t/(\text{ES}_t T\alpha)$  to the sum in  $Z_2$ .
  2. Run Algorithm 1 (next slide) for the three models:  $n$ ,  $\bar{e}$ ,  $t$ , bootstrap  $p$  and Student- $t$   $p$ .
  3. Run Algorithm 2 for the three models on the full sample and on the last 250 days:  $Z_2$ ,  $p$ , simulated 5% critical value.
  4. Plot the two null distributions of  $Z_2$  (GARCH-t) and the bootstrap distribution of  $t$ .
- **Report:** one table (model,  $n$ ,  $\bar{e}$ ,  $t$ , two  $p$ ,  $Z_2$ ,  $p$ , critical value, both samples) and one figure
- **Interpretation:** what does a rejection say about the joint forecasting procedure?
- **Notebook:** section B3

## B3 [Solved]: Two Algorithms — Task (2/2)

## Algorithm 1: McNeil–Frey residual bootstrap

1. Keep the breach days  $L_t > \text{VaR}_t$ ;  $e_t = (L_t - \text{ES}_t)/\sigma_t$ ,  $n$  residuals.
2.  $t = \bar{e}/(s_e/\sqrt{n})$ ,  $H_1: E[e_t] > 0$ .
3. Centre:  $\tilde{e}_t = e_t - \bar{e}$  (imposes  $H_0$ ).
4. 5000 times: draw  $n$  values of  $\tilde{e}$  with replacement (seed 0), compute  $t^*$ .
5.  $p = \text{share of } t^* \geq t$ ; assumes independent residuals.

Algorithm 2:  $Z_2$  under the model's own forecasts

1. Hold every  $\text{VaR}_t$ ,  $\text{ES}_t$  fixed;  $Z_2$  from the data.
2. 5000 times (seed 11), each day:  $U_t \sim U(0, 1)$ ; if  $U_t > 1 - \alpha$ ,  $L_t^* = F_t^{-1}(U_t)$ , else no breach.
3.  $F_t^{-1}$ : GARCH-t  $-\mu_t + \sigma_t t_\nu^{-1}(U)\sqrt{(\nu-2)/\nu}$ ; HS, FHS: the stored tail losses.
4.  $Z_2^* = 1 - \sum_t L_t^* 1\{U_t > 1 - \alpha\}/(\text{ES}_t T \alpha)$ .
5.  $p = \text{share } Z_2^* \leq Z_2$ ; critical value: 5% quantile of  $Z_2^*$ .

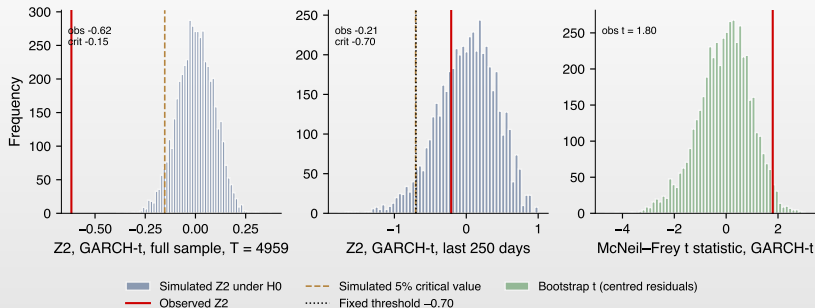
## B3 [Solved]: Solution (1/2) — Step by Step and in Full

- 12.03.2020:  $e_t = (9.99 - 13.43)/4.81 = -0.71$  (the loss was below the forecast ES); contribution  $0.744/123.98 = 0.0060$ , out of 192 breach days.

| Model   | $n$ | $\bar{e} / t$ | $p_{boot} / p_t$ | $Z_2 (T = 4\,959)$ | $p / \text{crit.}$ | $Z_2, p (250 \text{ d})$ |
|---------|-----|---------------|------------------|--------------------|--------------------|--------------------------|
| GARCH-t | 192 | 0.106 / 1.80  | 0.023 / 0.037    | -0.618             | <0.001 / -0.153    | -0.210, 0.289            |
| FHS     | 130 | 0.093 / 1.21  | 0.096 / 0.114    | -0.091             | 0.154 / -0.150     | 0.142, 0.602             |
| HS      | 156 | 0.363 / 2.88  | <0.001 / 0.002   | -0.405             | <0.001 / -0.154    | 0.723, 0.975             |

- GARCH-t:  $t = 1.80$ , bootstrap  $p = 0.023$  against Student- $t$   $p = 0.037$ : skewed residuals make the  $t$  approximation poor
- Full sample: GARCH-t ( $Z_2 = -0.618$ ) and HS rejected, FHS passes; last 250 days: nobody rejected, GARCH-t has only 6 breaches

## B3 [Solved]: Solution (2/2) — Null Distributions and Limits



- The null distribution of  $Z_2$  shrinks like  $1/\sqrt{T\alpha}$ : 5% critical value  $-0.153$  on 4 959 days,  $-0.705$  on 250 days (close to the reference  $-0.70$ )
- Interpretation: the GARCH-t rejection is evidence against the whole procedure (volatility filter, t tail, estimation); FHS, with the same filter, passes, which points to the tail but does not prove it; 5000 draws give a Monte Carlo s.e. of about 0.3 points at  $p = 5\%$

## B5 [Solved]: Is FHS Significantly Better? — Task

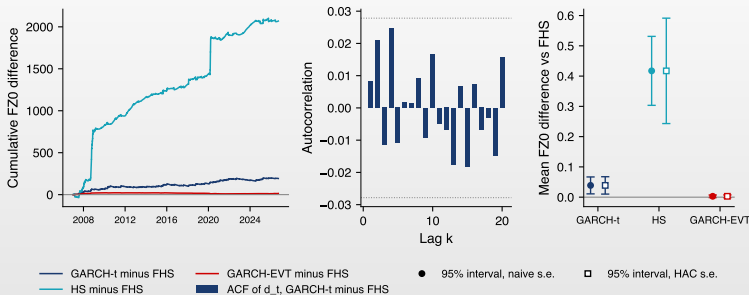
- **Question:** is FHS significantly better than GARCH-t, HS and GARCH-EVT under the FZ0 score on the S&P 500?
- **Inputs**
  - ▶ Score vectors `fz_losses('sp500')`: FZ0 of (VaR 2.5%, ES 2.5%) for the six models, aligned on the same 4959 days (Patton, Ziegel & Chen (2019))
  - ▶ Score differential  $d_t = S_{A,t} - S_{\text{FHS},t}$  (a difference of scores, not of losses); Newey–West with  $q = \lfloor 4(T/100)^{2/9} \rfloor = 9$  lags (Newey & West (1987))
- **Tasks**
  1. Compute the mean FZ0 of each model.
  2. For  $A = \text{GARCH-t}$ :  $\bar{d}$ , the autocovariances  $\gamma_0, \dots, \gamma_9$ ,  $\text{LRV} = \gamma_0 + 2 \sum_{k=1}^9 (1 - k/10) \gamma_k$ ,  $\text{se} = \sqrt{\text{LRV}/T}$ ,  $\text{DM} = \bar{d}/\text{se}$  and its  $p$ -value (Diebold & Mariano (1995)).
  3. Repeat for HS and GARCH-EVT; compute the naive  $t$  (ordinary s.e.) and the ratio  $\text{LRV}/\gamma_0$ .
  4. Plot the cumulative  $d_t$ , the ACF of  $d_t$  and the naive and HAC 95% intervals of  $\bar{d}$ .
  5. Optional: the Giacomini–White test with  $h_{t-1} = (1, d_{t-1})$  (Giacomini & White (2006)).
- **Report:** mean scores, one complete DM calculation, a DM table and one figure
- **Interpretation:** which differences are significant, and when does ignoring autocorrelation overstate significance?
- **Notebook:** section B5

## B5 [Solved]: Solution (1/2) — One DM Statistic Step by Step

1. Mean FZ0: FHS  $-3.551$ , GARCH-EVT  $-3.548$ , GARCH-t  $-3.512$ , HS  $-3.134$ , Student-t  $-3.036$ , Normal  $-2.958$ .
2. GARCH-t minus FHS:  $\bar{d} = 0.0388$ ;  $\gamma_0 = 1.0076$ ;  $\gamma_1, \dots, \gamma_9$ :  $0.0081, 0.0210, -0.0116, 0.0249, -0.0110, 0.0018, 0.0015, 0.0094, -0.0094$ .
3.  $2 \sum_k (1 - k/10) \gamma_k = 0.0551$ , so  $LRV = 1.0627$ ;  $se = \sqrt{1.0627/4959} = 0.0146$  (naive  $0.0143$ ).
4.  $DM = 0.0388/0.0146 = 2.65$ ,  $p = 0.008$ : GARCH-t is significantly worse at 5%.

| A minus FHS | $\bar{d}$ | DM   | $p$      | naive $t$ | $LRV/\gamma_0$ |
|-------------|-----------|------|----------|-----------|----------------|
| GARCH-t     | 0.0388    | 2.65 | 0.008    | 2.72      | 1.05           |
| HS          | 0.4173    | 4.70 | $<0.001$ | 7.18      | 2.34           |
| GARCH-EVT   | 0.0029    | 1.24 | 0.216    | 1.31      | 1.11           |

## B5 [Solved]: Solution (2/2) — Differentials, ACF and Intervals

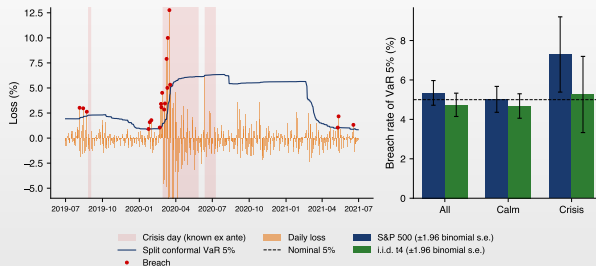


- HS loses steadily and in jumps (2008, 2020); its LRV is 2.34 times  $\gamma_0$ , so the naive  $t$  (7.18) overstates the HAC DM (4.70); for GARCH-t the ACF is small (lag 1: 0.008) and the two intervals almost coincide
- Provisional ranking: FHS and GARCH-EVT tie ( $p = 0.216$ ); GARCH-t and HS are significantly worse; GW (optional): GARCH-EVT 7.38 ( $p = 0.025$ ): the conditional tie fails, which moment drives it needs a separate test

## B7 [Solved]: Conformal VaR in Calm and Crisis Periods — Task

- **Question:** does split conformal VaR keep its coverage on crisis days of the S&P 500?
- **Inputs**
  - ▶ Timeline for forecast day  $t$  (all windows end at  $t - 1$ ): scale  $\hat{\sigma}_t = \text{sd of } r_{t-250}, \dots, r_{t-1}$ ; calibration scores  $s_{t-250}, \dots, s_{t-1}$ ,  $s_j = L_j / \hat{\sigma}_j$ ; crisis flag: sd of  $r_{t-20}, \dots, r_{t-1}$  above the 90th percentile of its values over the previous 1000 days
  - ▶  $\text{VaR}_t = \hat{\sigma}_t s_{(k)}$ ,  $k = \lceil 251(1 - \alpha) \rceil$ ; evaluation 03.01.2007–18.09.2026 (4 959 days); benchmark: i.i.d. Student- $t_4$  returns  $\times 1\%$ , seed 5, same length
- **Tasks**
  1. Build VaR 5% and VaR 1% and the breach series on the S&P 500.
  2. Fill the grid (data source  $\times$  level  $\times$  all/calm/crisis) with rates and counts; add the Kupiec  $p$  (all days) and the Christoffersen independence  $p$ .
  3. Repeat on the i.i.d. benchmark: exchangeable returns, rolling-standardised scores only approximately so.
  4. Plot a VaR path with crisis shading and the rates with their binomial intervals.
- **Report:** the grid, two  $p$ -values per series, one figure
- **Interpretation:** is the marginal guarantee met on real data, and what changes between real and exchangeable data: coverage or independence?
- **Notebook:** section B7

## B7 [Solved]: Solution



- S&P 500, VaR 5%: all  $265/4959 = 5.34\%$  (Kupiec  $p = 0.272$ ), calm  $213/4246 = 5.02\%$ , crisis  $52/713 = 7.29\%$ ; independence rejected ( $p < 0.001$ ); VaR 1%: 1.29% overall ( $p = 0.049$ ), 2.10% in crises
- i.i.d. benchmark: 4.74%, calm 4.68%, crisis 5.26% (27/513), independence not rejected ( $p = 0.965$ ); the binomial intervals ignore dependence, so the real-data ones are too narrow
- Interpretation: the average is close to 5%, but volatility clustering breaks exchangeability, so breaches cluster and crisis coverage fails

## C2 [Proposed]: Critique an AI Answer — Task

- **Question:** is the AI answer on the next slide correct, and does its conclusion survive once the code is fixed?
- **Model:** A3, A5
- **Inputs**
  - ▶ The prompt, the answer and its line-numbered code (next slide); `lr_ind(I)` is `christoffersen(I, 0.01) ['LR_ind']` from the notebook helpers
  - ▶ Fixture: 401 prices from 100, daily log returns  $0.01 \times t_4/\sqrt{2}$  (seed 8) with  $r = -8\%$  at position 320; real data: GSPC.INDX from 1990
- **Tasks**
  1. Find the three conceptual errors: in the forecast window, in the test distribution, in the regulatory reading.
  2. For each error, say how you would detect it: a check on the output, the lecture, a published table.
  3. Run the answer's code and your corrected code on the fixture and on the S&P 500; report VaR at position 320 and the exception counts.
  4. Give the correct zone, plus factor and multiplier for 7 exceptions (A5).
  5. Optional checks: the evaluation window of the traffic light;  $x = 0$  in the Kupiec formula.
- **Report:** the three errors with detection and correction, the corrected code, two exception counts
- **Interpretation:** does “the model must be withdrawn” survive?

## C2 [Proposed]: Critique an AI Answer — the Answer

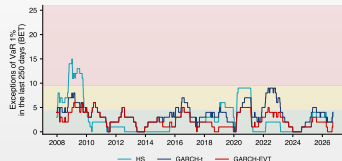
- **Prompt:** “I have daily S&P 500 closes in a pandas Series P. Write Python for a rolling 250-day historical-simulation VaR 1%, count the exceptions and run the Kupiec and Christoffersen tests. My last 250 days have 7 exceptions: what does that mean?”

```
1 import numpy as np, scipy.stats as stats
2 r = np.log(P).diff().dropna().values
3 W, p = 250, 0.01
4 VaR = np.array([-np.quantile(r[t-W:t+1], p) for t in range(W, len(r))])
5 I = (-r[W:] > VaR).astype(int)
6 T, x = len(I), I.sum(); ph = x / T
7 LR_uc = -2*((T-x)*np.log(1-p) + x*np.log(p) - (T-x)*np.log(1-ph) - x*np.log(ph))
8 LR_cc = LR_uc + lr_ind(I)
9 p_uc, p_cc = stats.chi2.sf(LR_uc, df=1), stats.chi2.sf(LR_cc, df=1)
```

- **Answer:** “Your last 250 days:  $x = 7$ ,  $LR_{uc} = 5.50$ ,  $p = 0.019$ : coverage rejected at 5%. Under the Basel traffic light 7 exceptions is the red zone, so the model must be withdrawn.”
- AI code usually runs without an error message: the mistakes are in the finance and the statistics, not in the syntax

## B2 [Proposed]: Traffic Light for GARCH-t on the BET — Task

- ▣ **Question:** would a supervisor have penalised the GARCH-t VaR 1% on the BET since 2007?
- ▣ **Model:** A5, B1
- ▣ **Inputs**
  - ▶ Panel `fc('bet')`, GARCH-t, columns `VaR1`, `VaR2.5`; evaluation 03.01.2007–18.09.2026 ( $T = 4937$ ); the first complete 250-day window ends on 21.12.2007, so there are 4688 windows; zones and multipliers: A5
- ▣ **Tasks**
  1. Count the VaR 1% exceptions in the last 250 days; give the zone,  $P(X \leq x)$  and the FRTB multiplier.
  2. For every window, the count and zone; report the share of windows in each zone and the worst window with its end date.
  3. Count the VaR 2.5% exceptions in the last 250 days and compare with the desk limit of 30.
  4. Compute the full-sample breach rate and Kupiec  $p$ -value.
- ▣ **Interpretation:** is time in yellow evidence against the model? (overlapping windows: shares are descriptive, not a test)
- ▣ **Notebook:** section B2



## Submission Checklist, Exit Question and Team Project

### Hand in before the next seminar

- A1, A3, A5, A7: the answers and the completed tables
- B1, B3, B5, B7: one table and one figure each, as in the solutions
- C2: errors, corrected code, two exception counts; B2: four numbers and the zone shares
- One interpretation sentence per exercise
- **Exit question:** which model would you provisionally choose for the S&P 500, on what evidence, and what is the main limitation of that evidence?

### Assessed team project (one)

- C1: a provisional VaR/ES recommendation for a Romanian trading book (extension pack 6), as a Quantlet folder in the team repository
- Rubric: data and timing correct (no look-ahead), minimum test set, evidence table, recommendation with its limitation
- Report actual AI use in AI\_USE.md and every error you found in AI\_ERRORS.md: request, answer, error, how found, correction; no minimum number of errors
- Extension packs are optional

## Extension Packs: What Each One Needs

| Pack                    | Exercises                    | Prerequisites           |
|-------------------------|------------------------------|-------------------------|
| 1. Test theory          | A1 Extended [Solved], A2, A4 | A1, A3                  |
| 2. Scores and ES theory | A6 [Solved], A9, A10         | A7, B5                  |
| 3. Conformal VaR        | A8 [Solved], B8              | B7                      |
| 4. More markets         | B4, B6                       | B3, B5                  |
| 5. Estimation risk      | B9                           | B1, Lecture 8 Section 2 |
| 6. Projects             | C1 (assessed), C3            | B1–B5; C3 also needs R  |

- Each pack has its own notebook section; do them in order within a pack
- C1 and C3 are possible starts for the team project

## A1 Extended [Solved]: Why $LR_{uc}$ Is $\chi_1^2$ — Task

- ▣ **Question:** where do the  $\chi_1^2$  limit and the noncentrality  $\lambda$  of A1 come from?
- ▣ **Inputs:** i.i.d. Bernoulli( $\pi$ ) hits,  $\ell(\pi) = x \ln \pi + (T - x) \ln(1 - \pi)$ ,  $LR_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)]$ ; local alternatives  $\pi_T = \alpha + h/\sqrt{T}$  with  $h$  fixed
- ▣ **Tasks**
  1. Expand  $\ell(\alpha)$  to second order around  $\hat{\pi}$  and show  $LR_{uc} = T(\hat{\pi} - \alpha)^2/(\alpha(1 - \alpha)) + o_p(1)$  under  $H_0$ .
  2. Use the CLT to conclude  $LR_{uc} \xrightarrow{d} \chi_1^2$ .
  3. Under  $\pi_T$ , derive the limit  $\chi_1^2(\lambda)$  and give  $\lambda$ .
- ▣ **Report:** the three steps
- ▣ **Interpretation:** why is the local approximation poor at  $\alpha T = 2.5$ ?
- ▣ Notebook: section A1

## A1 Extended [Solved]: Solution

1.  $\ell'(\hat{\pi}) = 0$  at the MLE;  $\ell(\alpha) = \ell(\hat{\pi}) + \frac{1}{2}\ell''(\bar{\pi})(\alpha - \hat{\pi})^2$  with  $\bar{\pi}$  between  $\alpha$  and  $\hat{\pi}$ , and  $-\ell''(\bar{\pi})/T = \frac{\hat{\pi}}{\bar{\pi}^2} + \frac{1-\hat{\pi}}{(1-\bar{\pi})^2} \xrightarrow{P} \frac{1}{\alpha(1-\alpha)}$ .
  2. So  $\text{LR}_{uc} = Z_T^2 + o_p(1)$ ,  $Z_T = \sqrt{T}(\hat{\pi} - \alpha)/\sqrt{\alpha(1-\alpha)} \xrightarrow{d} N(0, 1)$  by the CLT:  $\chi_1^2$ , critical value 3.84.
  3. Under  $\pi_T$ :  $\sqrt{T}(\hat{\pi} - \alpha) = \sqrt{T}(\hat{\pi} - \pi_T) + h$ , so  $Z_T \xrightarrow{d} N(h/\sqrt{\alpha(1-\alpha)}, 1)$  and  $\text{LR}_{uc} \xrightarrow{d} \chi_1^2(\lambda)$ ,  $\lambda = h^2/(\alpha(1-\alpha)) = T(\pi - \alpha)^2/(\alpha(1-\alpha))$ .
- Interpretation: the expansion needs many breaches; with 2.5 expected the distribution of  $x$  is a coarse lattice, hence the gap between 36% (local) and 24% (exact) in A1

## A2 [Proposed]: How Many Days to Detect a Doubled Rate? — Task

- ▣ **Question:** how many days are needed to detect, with 80% power, a VaR 2.5% whose true breach rate is 5%?
- ▣ **Model:** A1
- ▣ **Inputs:**  $\alpha = 0.025$ ,  $\pi = 0.05$ , the rule “reject if  $LR_{uc} > 3.84$ ”; the FRTB desk limit of 30 exceptions of VaR 2.5% is a separate regulatory rule
- ▣ **Tasks**
  1. At  $T = 250$ :  $\lambda$ , the local power, the exact acceptance region, the exact power and the actual size.
  2. Find  $\lambda^*$  with  $P(\chi_1^2(\lambda^*) > 3.84) = 0.80$  by root finding and solve for the asymptotic  $T$ .
  3. For  $T = 100, \dots, 700$ : acceptance set  $\{x : LR_{uc}(x) \leq 3.84\}$ , power =  $1 - P(X \in \text{set} \mid \pi)$ ; plot power against  $T$  and find the first  $T$  with power  $\geq 80\%$ .
  4. Check whether the power stays above 80% for every larger  $T$ .
  5. Compute  $P(X > 30)$  for  $X \sim \text{Binomial}(250, 0.05)$ .
- ▣ **Report:**  $\lambda$ , powers, size,  $\lambda^*$ , the two values of  $T$ ,  $P(X > 30)$ , one figure
- ▣ **Interpretation:** what does the desk limit screen for?
- ▣ **Notebook:** section A2

## A4 [Proposed]: LR versus DQ at $x = 7$ — Task

- **Question:** why do  $LR_{uc}$  and the intercept-only DQ statistic give different numbers for the same breaches?
- **Model:** A3
- **Inputs**
  - ▶ Hit:  $T \times 1$  vector of  $I_t - \alpha$ ;  $X$ :  $T \times k$  regressors known at  $t - 1$ , full rank;  $\text{Hit} = X\beta + u$ ;  $Q = \text{plim } X'X/T$  positive definite ([Engle & Manganelli \(2004\)](#))
  - ▶ Example design with one lag: rows  $X_t = (1, \text{Hit}_{t-1})$ , so  $k = 2$
- **Tasks**
  1. Show  $E[\text{Hit}_t \mid \mathcal{F}_{t-1}] = 0$  and  $\text{Var}(\text{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$  under  $H_0$ .
  2. With  $X_t = 1$  show  $\text{DQ} = T(\hat{\pi} - \alpha)^2 / (\alpha(1 - \alpha))$ ; compute it for  $T = 250$ ,  $x = 7$  and compare with  $LR_{uc} = 5.50$  ( $p = 0.019$ ).
  3. Compute both statistics for  $x = 0, \dots, 15$  and their acceptance sets at 3.84.
  4. Advanced: derive  $\text{DQ} \xrightarrow{d} \chi_k^2$  from a martingale-difference CLT for  $T^{-1/2}X'\text{Hit}$ .
- **Report:** the two statistics and  $p$ -values, the acceptance sets, the derivation
- **Interpretation:** do the two tests disagree in their decision or only in their numbers?
- **Notebook:** section A4

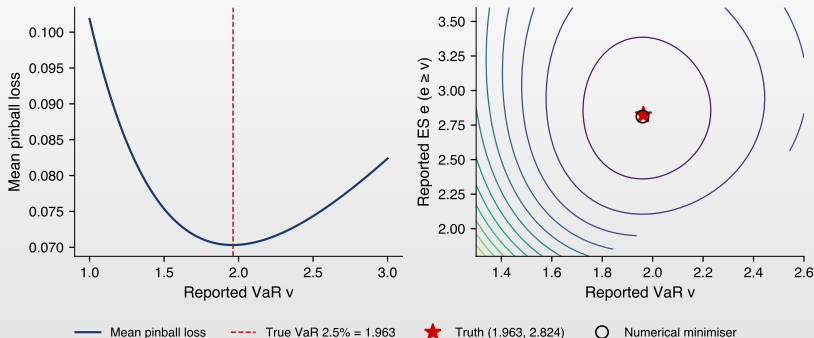
## A6 [Solved]: Consistency of the Pinball and FZ0 Losses — Task

- ▣ **Question:** why does the truth minimise the expected pinball and FZ0 scores, and does a simulation confirm it?
- ▣ **Inputs**
  - ▶  $L$  with  $E|L| < \infty$ , density  $f > 0$  near  $\text{VaR}_\alpha$ ,  $\text{ES}_\alpha > 0$ ; forecasts  $e \geq v > 0$ ;  $\alpha = 2.5\%$
  - ▶ Simulation:  $L = \sqrt{(4-2)/4} Z$ ,  $Z \sim t_4$ ,  $10^6$  draws, seed 5; truth  $\text{VaR} = 1.963$ ,  $\text{ES} = 2.824$  (unit-variance  $t_4$  formulas)
- ▣ **Tasks**
  1. Differentiate  $E[(1\{L > v\} - \alpha)(L - v)]$  in  $v$ , find the zero and show it is a minimum.
  2. For  $\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$  compute  $\partial_v$  and  $\partial_e$  of the expectation and show that the true pair is the unique minimiser.
  3. Minimise the average pinball and FZ0 losses on the simulated draws and plot the score curve and the FZ0 contours.
- ▣ **Report:** derivatives, the argument for the minimum, numerical minimisers, one figure
- ▣ **Interpretation:** why does consistency make ranking by average score meaningful?
- ▣ **Notebook:** section A6

## A6 [Solved]: Solution (1/2) — Derivations

1.  $\frac{d}{dv} E[(1\{L > v\} - \alpha)(L - v)] = F(v) - (1 - \alpha)$ : negative below and positive above  $\text{VaR}_\alpha$ , zero at  $F(v) = 1 - \alpha$ ; second derivative  $f(v) > 0$ : a unique minimum.
2.  $\partial_v E[\text{FZ0}] = \frac{1}{e} (1 - \frac{P(L > v)}{\alpha})$ : same sign pattern, zero only at  $v = \text{VaR}_\alpha$ , for every  $e$ .
3. At that  $v$ ,  $\partial_e E[\text{FZ0}] = \frac{1}{e^2} (e - v - E[(L - v)1\{L > v\}]/\alpha) = \frac{1}{e^2} (e - \text{ES}_\alpha)$ : negative for  $e < \text{ES}_\alpha$ , positive above: the profiled objective in  $e$  has its unique minimum at  $\text{ES}_\alpha$ .
4. Identification functions:  $V_1 = \alpha - 1\{L > v\}$ ,  $V_2 = e - v - \frac{1}{\alpha} 1\{L > v\}(L - v)$ ; their means vanish only at the true pair (Fissler & Ziegel (2016)).

## A6 [Solved]: Solution (2/2) — Numerical Check



- Minimisers: pinball  $v = 1.960$ ; FZ0  $(v, e) = (1.960, 2.812)$ ; truth (1.963, 2.824): equal up to simulation error
- Interpretation: a forecaster who knows  $F$  minimises the expected score by reporting the truth, so a lower average score is evidence of a better forecast

## A9 [Proposed]: ES Alone Is Not Elicitable — Task

- ▣ **Question:** why can no score of ES alone reward honest forecasts?
- ▣ **Model:** A6
- ▣ **Inputs**
  - ▶  $T(F)$  is elicitable if  $T(F) = \arg \min_x E_F[S(x, L)]$  with a unique minimiser ([Gneiting \(2011\)](#));  $\delta_c$ : the point mass at  $c$  (loss  $c$  for sure)
  - ▶ ES for any distribution, atoms included:  $ES_\alpha = \frac{1}{\alpha} \int_{1-\alpha}^1 q_u du$ , the mean of the upper- $\alpha$  tail quantiles of the loss (an atom may be split)
- ▣ **Tasks**
  1. Show that if  $T$  is elicitable, every level set  $\{F : T(F) = t\}$  is closed under mixtures  $(1 - \lambda)F_0 + \lambda F_1$ .
  2.  $F_0 = \delta_1$ ;  $F_1$ :  $P(L = 2) = \alpha/2$ ,  $P(L = 0) = 1 - \alpha/2$ : compute  $ES_\alpha$  of  $F_0$ ,  $F_1$  and  $\frac{1}{2}F_0 + \frac{1}{2}F_1$  from the tail-quantile picture.
  3. Explain, conceptually, why this does not contradict the joint elicibility of (VaR, ES).
- ▣ **Report:** the argument, three ES values, one sentence
- ▣ **Interpretation:** what does this imply for ranking ES forecasts?
- ▣ **Notebook:** section A9

## A10 [Proposed]: Moments of the Cumulative Violation — Task

- **Question:** what are the null moments of  $H_t$ , and is the Normal critical value adequate at  $T = 250$ ?
- **Model:** A7
- **Inputs:** PITs  $u_t$  i.i.d.  $U(0, 1)$  under a correct forecast;  $H_t = \frac{1}{\alpha}(u_t - (1 - \alpha))1\{u_t > 1 - \alpha\}$  (Du & Escanciano (2017)): zero on 97.5% of days, rising linearly to 1 in the tail;  $\alpha = 2.5\%$
- **Tasks**
  1. Plot  $H(u)$  for  $u \in (0.9, 1)$ .
  2. Compute  $E[H]$  and  $E[H^2]$  by integrating over  $(1 - \alpha, 1)$ ; deduce  $\text{Var}(H) = \alpha(1/3 - \alpha/4)$ .
  3. Check both moments on  $4 \cdot 10^6$  uniform draws (seed 6).
  4. Compute the one-sided Normal 5% cut-off for  $H$  at  $T = 250$ ; simulate 20 000 samples of 250 PITs (seed 12) and compare with their 95th percentile.
- **Report:** the moments, the check, both cut-offs and the actual size of the Normal cut-off
- **Interpretation:** why does  $H_t$  carry more information than  $I_t$ ?
- **Notebook:** section A10

## A8 [Solved]: Conformal Quantile and One ACI Step — Task

□ **Question:** how does split conformal VaR 10% react to one breach under ACI?

□ **Inputs**

- ▶  $n = 19$  calibration scores (sorted, table); tomorrow's  $\hat{\sigma} = 1.4\%$ ; target  $\alpha = 0.10$  (fixed); ACI step  $\gamma = 0.02$ ,  $\alpha_1 = 0.10$  (Gibbs & Candès (2021))
- ▶ For this one-step illustration the scores and  $\hat{\sigma}$  are frozen; if  $k > n$  the quantile is  $+\infty$

□ **Tasks**

1. Compute  $k = \lceil (n+1)(1-\alpha) \rceil$ ,  $s_{(k)}$  and  $\text{VaR} = \hat{\sigma}s_{(k)}$ .
2. Compute  $\alpha_2 = \alpha_1 + \gamma(\alpha - \text{err}_1)$  after a breach and after no breach.
3. Recompute  $k$  and VaR for both values of  $\alpha_2$ ; below which  $\alpha_t$  does the VaR become infinite?

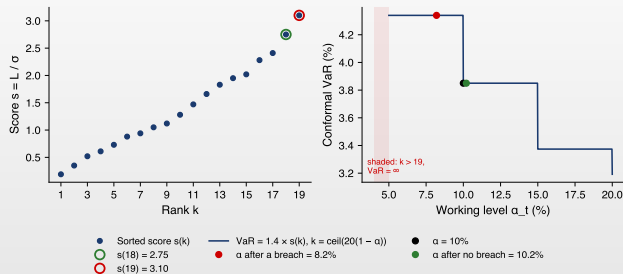
□ **Report:**  $k$ , three VaR values, the infinite-VaR threshold

□ **Interpretation:** how smoothly does ACI move the VaR?

□ **Notebook:** section A8

| $k$ | $s_{(k)}$ | $k$ | $s_{(k)}$ | $k$ | $s_{(k)}$ | $k$ | $s_{(k)}$ |
|-----|-----------|-----|-----------|-----|-----------|-----|-----------|
| 1   | 0.19      | 6   | 0.88      | 11  | 1.47      | 16  | 2.28      |
| 2   | 0.35      | 7   | 0.94      | 12  | 1.66      | 17  | 2.41      |
| 3   | 0.52      | 8   | 1.05      | 13  | 1.83      | 18  | 2.75      |
| 4   | 0.61      | 9   | 1.12      | 14  | 1.95      | 19  | 3.10      |
| 5   | 0.73      | 10  | 1.28      | 15  | 2.02      |     |           |

## A8 [Solved]: Solution



- $k = \lceil 20 \cdot 0.9 \rceil = 18$ ;  $s_{(18)} = 2.75$ ;  $\text{VaR} = 1.4 \cdot 2.75 = 3.85\%$
- After a breach:  $\alpha_2 = 0.10 + 0.02(0.10 - 1) = 0.082$ ,  $k = \lceil 18.36 \rceil = 19$ ,  $\text{VaR} = 1.4 \cdot 3.10 = 4.34\%$
- No breach:  $\alpha_2 = 0.102$ ,  $k = \lceil 17.96 \rceil = 18$ : the VaR does not move; for  $\alpha_t < 1/(n+1) = 5\%$ ,  $k = 20 > n$ :  $\text{VaR} = \infty$
- Interpretation: with small  $n$  ACI moves in jumps of one order statistic; small changes of  $\alpha_t$  are invisible until they cross a grid point  $j/(n+1)$

## B8 [Proposed]: Conformal VaR on the BET and Bitcoin — Task

- **Question:** does ACI repair crisis coverage on the BET and on Bitcoin, or only the long-run average?
- **Model:** B7
- **Inputs**
  - ▶ BET (BET.csv) from 03.01.2007 and Bitcoin (BTC-USD.CC.csv) from 14.06.2017, to 18.09.2026; VaR 5%: split conformal, ACI and GARCH-t; scores, windows and crisis flag as in B7
  - ▶ ACI:  $\alpha_1 = 0.05$ ,  $\gamma = 0.005$ , started on the first evaluation day; an infinite VaR counts as no breach and is reported separately
- **Tasks**
  1. For each market and method, report breach counts and rates on all, calm and crisis days.
  2. Report the number of days with an infinite ACI VaR.
  3. Attach a binomial 95% interval to each crisis rate and say which differences from 5% are clear.
- **Report:** a rate-and-count table, the infinite-VaR count, one figure
- **Interpretation:** an evidence-qualified comparison of the three methods, not a single winner
- **Notebook:** section B8

## B4 [Proposed]: $Z_2$ by Simulation on the BET and Bitcoin — Task

- ▣ **Question:** are the ES 2.5% forecasts of GARCH-t and GARCH-EVT calibrated on the BET and on Bitcoin?
- ▣ **Model:** B3
- ▣ **Inputs**
  - ▶ Panels `fc('bet')` (from 2007) and `fc('btc')` (from 2017); Algorithm 2 of B3 with 5000 draws, seed 11, forecasts held fixed
  - ▶ Samplers: GARCH-t as in B3; GARCH-EVT  $L^* = -\mu_t + \sigma_t \left( u + \frac{\beta}{\xi} \left[ \left( \frac{1-u}{p_u} \right)^{-\xi} - 1 \right] \right)$  with the stored  $u, \xi, \beta, p_u$
- ▣ **Tasks**
  1. Fill the four-row template: market, model,  $Z_2$ ,  $p$ , 5% critical value, breach days, on the full sample.
  2. Repeat on the last 250 days.
  3. Plot the four null distributions with the observed  $Z_2$ .
- ▣ **Report:** the eight-row table and one figure
- ▣ **Interpretation:** a provisional comparison with its limitations: why does the verdict change between the full sample and the last year?
- ▣ **Notebook:** section B4

## B6 [Proposed]: Comparing Models on Bitcoin and EUR/RON — Task

- ▣ **Question:** can the data separate the six models on Bitcoin and on EUR/RON?
- ▣ **Model:** B5
- ▣ **Inputs**
  - ▶ FZ0 score vectors `fz_losses('btc')`, `fz_losses('eurron')`; DM with Newey–West lags  $\lfloor 4(T/100)^{2/9} \rfloor$
  - ▶ MCS ([Hansen, Lunde & Nason \(2011\)](#)):  $T_{max}$  statistic, circular block bootstrap, blocks of 10 days, 1000 replications, seed 2, 90% set
  - ▶ Murphy diagram ([Ehm, Gneiting, Jordan & Krüger \(2016\)](#)): Bitcoin VaR 2.5%, HS versus GARCH-EVT, 240 thresholds from 2.8% to 17.2%, pointwise HAC bands
- ▣ **Tasks**
  1. Find the lowest mean FZ0 in each market and run exploratory DM tests of every other model against it.
  2. Run the MCS; show the first elimination step (statistic,  $p$ -value, model removed) and list the 90% set.
  3. Plot the Murphy difference with its bands and say whether the curves cross.
- ▣ **Report:** a DM/MCS table, the elimination steps, one figure
- ▣ **Interpretation:** which models remain in the MCS for each market, and what limits the separation?
- ▣ **Notebook:** section B6

## B9 [Proposed]: Kupiec Size Under Estimation Risk — Task

- ▣ **Question:** how often does Kupiec reject a correct GARCH-t model when its parameters are estimated?
- ▣ **Model:** B1
- ▣ **Inputs**
  - ▶ Frozen DGP: GARCH(1,1)-t fitted to the S&P 500 1990–2026 ( $a = 0.102$ ,  $b = 0.892$ ,  $\nu = 6.1$ ); 400 paths of  $R + 5000$  days after a burn-in of 500, seed 9
  - ▶ Fixed scheme: estimate once on the first  $R = 1000$  days ( $\nu$  floored at 2.2), then forecast VaR 1% for  $P = 250, 1000, 5000$  days; on the same path also the true-parameter VaR
- ▣ **Tasks**
  1. Record the Kupiec rejection at 5% for the estimated and the true forecasts, paired on each path.
  2. Report each rejection rate with its own Monte Carlo s.e.  $\sqrt{\hat{p}(1 - \hat{p})/400}$ .
  3. Compare the sd of the breach rate across paths with  $\sqrt{\alpha(1 - \alpha)/P}$ ; plot the rates against  $P/R$ .
  4. Going further: an actual bootstrap calibration: simulate under the fitted null, re-estimate on each replicate and take the 95th percentile of  $LR_{uc}$  as the critical value.
- ▣ **Report:** a table ( $P$ , two rates with s.e., two sd) and one figure
- ▣ **Interpretation:** how does the distortion grow with  $P/R$  (Escanciano & Olmo (2010))?
- ▣ **Notebook:** section B9

## C1 [Proposed]: A Provisional Model for a Romanian Trading Book — Task (1/2)

- **Question:** which VaR/ES model would you provisionally recommend for a book in lei, half in the BET and half in euro?
- **Inputs**
  - ▶ BET closes (BET.csv) and the BNR EUR/RON reference rate; join the **prices** on common days, then simple returns, then  $r_t^p = \ln(1 + \frac{1}{2} R_t^{BET} + \frac{1}{2} R_t^{EUR})$ , daily rebalanced, no costs
  - ▶ The six models of Lecture 8, 1000-day window; evaluation from 09.07.2009 (after 2008: the 2008 crisis is an unobserved stress scenario) to 18.09.2026
- **Tasks**
  1. Build the portfolio and plot its losses with the VaR 1% of HS, FHS and GARCH-EVT.
  2. Run the minimum test set (next slide) for all six models.
- Notebook: section Part C

## C1 [Proposed]: Evidence and Recommendation — Task (2/2)

- **Question:** what does the evidence support, and what does it not?
- **Model:** B1–B5
- **Inputs:** minimum test set: frequency (Kupiec, CC), timing (durations or DQ), ES ( $Z_2$  by simulation), comparison (FZ0, DM, 90% MCS), regulation (worst and last 250-day windows, desk limits)
- **Tasks**
  1. Fill the evidence table: model  $\times$  rate,  $p_{cc}$ ,  $p_{dur}$ ,  $Z_2$  /  $p$ , worst window, last window (VaR 1% / VaR 2.5%), FZ0 / MCS  $p$ .
  2. State the size and power of the battery at  $T = 250$  (B1, A1).
  3. Write one provisional recommendation with a challenger model, and list what FRTB approval would also require (desk P&L backtests, P&L attribution).
- **Report:** a Quantlet folder and one slide: figure, evidence table, recommendation, main limitation
- **Interpretation:** would the choice survive a year like 2008 or 2020?
- **Notebook:** section Part C

## C3 [Proposed]: Backtesting ES with ES Forecasts Only — Task (1/2)

- **Question:** do the regression-based ES backtests of [Bayer & Dimitriadis \(2022\)](#) agree with  $Z_2$  and McNeil–Frey on the Lecture 8 forecasts?
- **Inputs**
  - ▶ The ES 2.5% forecasts of the six models in the four markets (24 series); returns convention of the paper:  $r_t = -L_t$ ,  $\hat{e}_t = -\widehat{ES}_t < 0$
  - ▶ R (4.x) with the authors' package: `esback::esr_backtest(r, e, alpha = 0.025, version = v)`;  $v = 1$ : Strict ESR (two-sided),  $v = 3$ : Intercept ESR (one-sided  $H_0: \gamma_1 \geq 0$ ); misspecification-robust covariance (default)
- **Tasks**
  1. Work one case first: S&P 500, GARCH-t: print the coefficients, their standard errors, the statistic and the  $p$ -value of both versions.
  2. Run both versions on all 24 series and draw a market-by-model heat map of the  $p$ -values.
  3. Compare the decisions at 5% with  $Z_2$  and McNeil–Frey (Lecture 8, Section 4).

## C3 [Proposed]: Blocks and Size — Task (2/2)

- **Question:** how do the tests behave on one-year blocks, and what is their size at  $T = 250$ ?
- **Model:** B3
- **Inputs:** non-overlapping 250-day blocks of the S&P 500 from the first evaluation day (drop the incomplete last block); the EGARCH(1,1)-t process of Eq. 2.1 of the paper, 1000 paths, a fixed seed
- **Tasks**
  1. Report the share of blocks in which each model is rejected, and the number of failed fits (excluded from the share).
  2. Simulate 1000 paths of 250 days, backtest the true ES 2.5% with the Intercept ESR, two-sided and one-sided, and report the rejection rates with their Monte Carlo s.e.
  3. Compare the two-sided rate with the 0.04 of Table 1 (robust covariance) and the one-sided rate with the nominal 5% (Table 2 reports it only at  $T = 2500$ ).
- **Report:** a Quantlet folder and two slides: the decision heat map with  $Z_2$  and McNeil–Frey, and the block shares with the size check
- **Interpretation:** when ESR and  $Z_2$  disagree, which would you trust as a supervisor who receives only ES forecasts, and why?

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